

Supplementary Material

Individual organization remains detectable across acute perturbations in bats

This supplement contains source/provenance tables, frozen inference definitions, descriptive developmental decompositions, exact-null resolution context, and the evidence-tier rules used for the comparative synthesis.

Supplementary Table S1. Public source and analysis provenance

Source	Species	Public archive	Programme role	Biological n	Source-level status
Harten et al. 2020	Rousettus aegyptiacus	Mendeley 10.17632/n9d8gbz3xr.1	first-flight history formation/ maintenance	14	primary formation FAIL; late secondary supported
Rachum et al. 2025	Rousettus aegyptiacus	Mendeley 10.17632/wh7c636y3t.1	randomized enrichment / individualization	29	unsupported individualization
Elie et al. 2024	Rousettus aegyptiacus	Mendeley 10.17632/h5ff9vv5pc.1	randomized auditory-feedback development	10	no difference in total individualization
Taub & Yovel 2020	Pipistrellus kuhlii	source public archive / source paper	external sensory masker	6	supported
Foskolos et al. 2022	Myotis daubentonii	Zenodo 10.5281/zenodo.4946256; Dryad 10.5061/dryad.ngf1vhhv3	graded masking	3 complete	supported; small n
Diebold et al. 2024	Eptesicus fuscus	Zenodo 10.5281/zenodo.13857870	reversible auditory-midbrain perturbation	4	supported; small n
Aharon et al. 2017	Pipistrellus kuhlii	Mendeley 10.17632/f6mvhj5gj9.3	navigation-context manipulation	4	supported
Teshima et al. 2026	Rhinolophus nippon	source public archive	coarse policy portability	5 adult mechanism cohort	supported; small n
Eveland et al. 2026	Carollia perspicillata	public repository 00keveland/ Tunnel_2026	external fixed-representation transfer	fixed evaluable cohort	coarse supported; detailed geometry unsupported
frozen wild carrier programme	multiple wild bat panels	source Movebank datasets	laboratory-to-wild boundary	4 panels	2/4 FAIL

Supplementary Table S2. Frozen inference definitions

Source / analysis	Biological unit	Frozen endpoint / representation	Scale / shared-context removal	Biological support	Frozen null	Primary rule / status
Harten first-flight primary	juvenile bat	individual Spearman slope (B_i) of own-history predictive advantage vs prior valid-day count	source-coordinate spatial-history score; equal individual weighting	14 juveniles, target valid-day ordinals 3–20	9,999 whole-history identity permutations within cohort	programme excess >0, p<=0.05, >=70% positive slopes; FAIL because 8/14 positive
Harten late-history secondary	juvenile bat	mean late own-history advantage (L_i), target ordinals 11–20	same frozen source-coordinate estimator	same 14 juveniles	same frozen permutation architecture	predeclared secondary; L=+112.5554, p=0.0001, 12/14 positive

Source / analysis	Biological unit	Frozen endpoint / representation	Scale / shared-context removal	Biological support	Frozen null	Primary rule / status
Harten recent-vs-earliest diagnostic	juvenile bat	(Q): recent two-day history advantage minus earliest two-day history advantage	equal-sized two-day history windows	same 14 juveniles	post-primary diagnostic permutation	diagnostic only ; cannot rescue failed primary
Rachum laboratory individualization	bat	3-D change vector: Boldness, Exploration, Activity	traits standardized from pooled Trials 1–2 only; treatment-group mean change removed	Season-2 complete n=29, 14 enriched / 15 impoverished	199,999 origin-stratified random assignments, seed 202610070817	D=V_enriched-V_impoverished >0 and p<=0.05; unsupported , p=0.167785
Elie developmental auditory feedback	bat	28-D whole-repertoire adult vocal centroid	treatment-blind feature scaling; sex×treatment centroid removed	10 randomized bats, 5 hearing / 5 deafened; >=1,037 finite calls per bat×feature	exact 120 sex-conditioned assignments	two-sided treatment contrast in total residual dispersion; no difference in amount , p=0.716667
Taub & Yovel masker	bat	source-native approach/ movement angle personal-bias state	condition-centered personal bias	6 bats baseline → masker; independent foam contrast 5 bats	exact complete identity permutations: 6!=720 and 5!=120	K>0 with exact upper-tail support; both contrasts supported
Foskolos graded masking	bat	log flight-time identity state across five noise levels	shared condition mean removed	3 bats complete across all 5 levels	exact 1,296 legal condition-wise identity assignments	positive programme identity advantage + exact support; supported, small n
Diebold auditory-midbrain perturbation	bat	4-D trial vocal vector: duration, bandwidth, IPI, call rate	treatment×trialtype pooled mean removed without identity; pooled residual SD	4 DREADD bats with >=5 saline and >=5 ligand trials	exact 4!=24 saline/ligand identity mappings	K>0, p<=0.05; supported , true mapping rank 1/24
Aharon navigation-context manipulation	bat	bilateral turning-location vector from per-trial left/right medians	source-condition mean removed across 4 bats	4 bats × 3 source conditions; >=10 valid trials/cell	con anchored; labels independently permuted in 75 and 300: (4!)=576	K>0, p<=0.05; supported , rank 2/576
Teshima/Rhino transparent policy	bat	fixed (I,M) from eight 3-D trajectory features	feature z-scoring within obstacle environment	5 adult <i>Rhinolophus nippon</i>	9,999 environment-wise identity permutations, frozen seed	K>0, p<=0.05, positive individual fraction gate; supported , 5/5 positive
Eveland/Carollia fixed I/M	bat	Rhino-fixed (I,M) representation	feature standardization within fixed date block	7 evaluable bats across two date blocks	9,999 trial-label shuffles within date block, seed 202610051141	K>0, p<=0.05, >=70% positive bats, both block means >0; supported
Eveland/Carollia detailed geometry	bat	fixed scale-free Rhino trajectory-geometry representation	same fixed date-block architecture	same 7-bat external cohort	9,999 fixed-architecture permutations	unsupported , K=-0.0350, p=0.2144
Wild scalar carrier bridge	panel	frozen scalar FlightIntensity persistence	source-panel-specific frozen estimator	4 wild panels	panel-level frozen PASS/FAIL gates	bridge opens only if >=3/4 panels support; observed 2/4 -> FAILED FROZEN GATE
Wild H/V localization	selected wild panel	post-outcome H/V components	harmonized/post-outcome representation	selected contexts only	post-outcome diagnostics	exploratory only ; cannot reopen scalar bridge

Supplementary Figure S1. Analysis-provenance timeline

The provenance timeline summarizes the sequence:

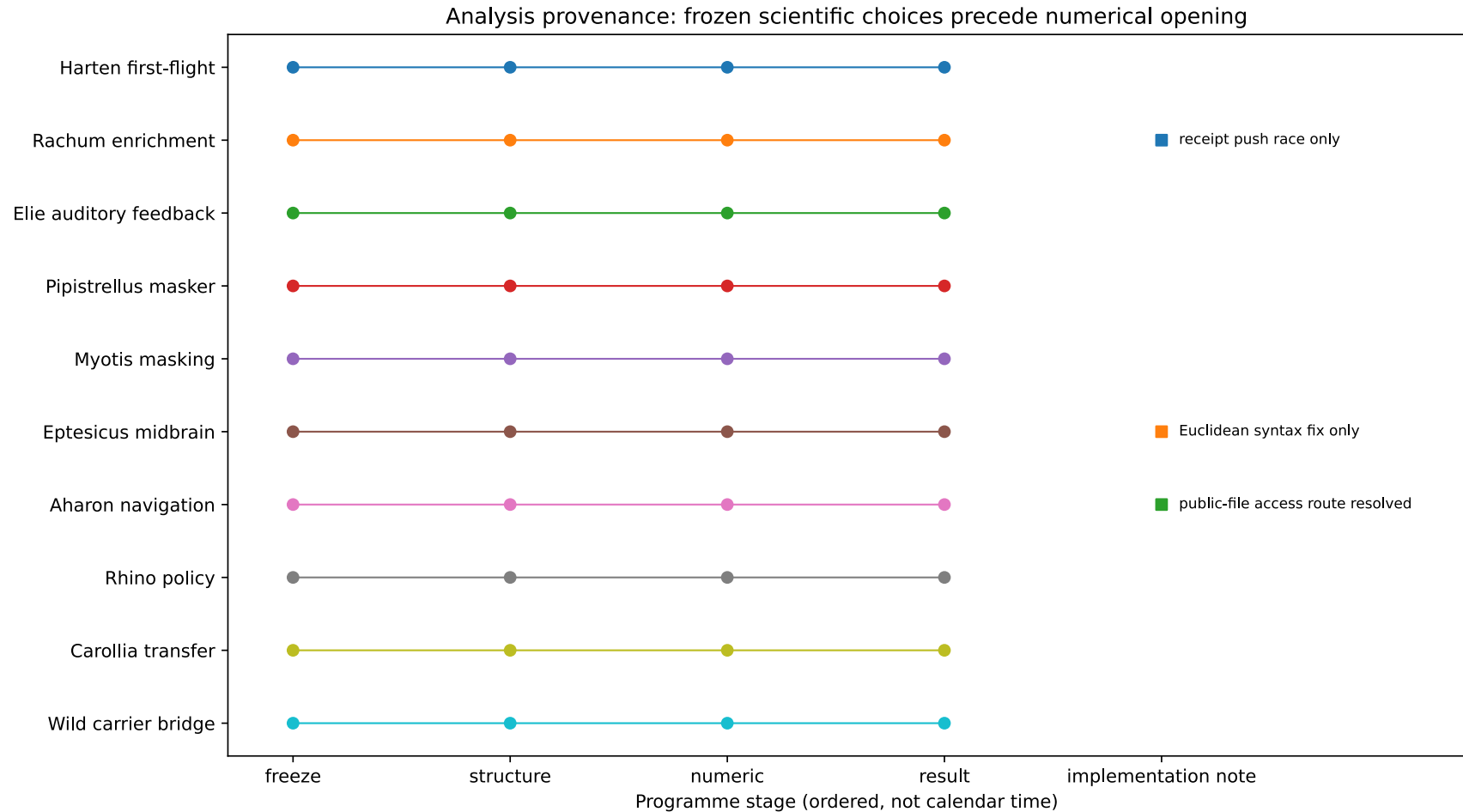
1. endpoint / support / null frozen;
2. structure or schema opened;
3. numerical outcome opened;

4. frozen result recorded;
5. implementation-only correction, when one occurred.

Special cases shown explicitly: - Eptesicus: a Euclidean-distance syntax error was corrected without changing the frozen endpoint, statistic or null before the successful rerun; - Aharon: an initial API-access failure was resolved through the anonymous public file route before the frozen Figure-1 primary opened; - Rachum: the primary calculation completed reproducibly while an output-receipt push race affected persistence only, not the numerical result.

No scientific endpoint was changed by these implementation events.

See Supplementary Material Figure S1.



Supplementary Material Figure S1. Analysis-provenance timeline.

Supplementary Figure S2. Developmental descriptive decompositions

Rachum enrichment

All three frozen trait contributions to the unsupported primary D are shown:

- Boldness: +0.465792;
- Exploration: +0.250837;
- Activity: +0.018070.

All three point in the enriched > impoverished direction. No trait-wise p-values are calculated.

Elie developmental auditory feedback

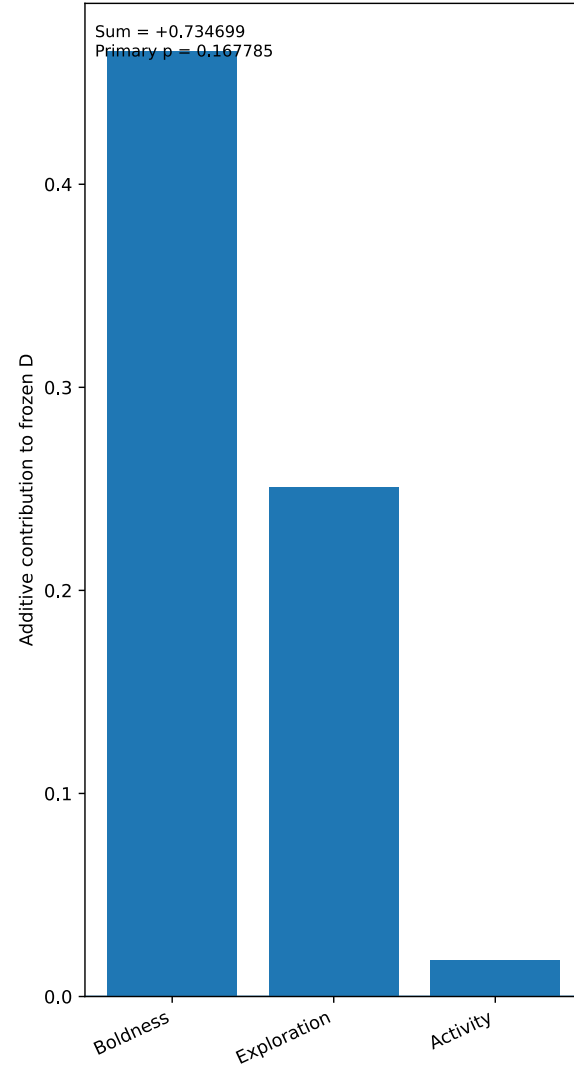
All 28 feature contributions to the frozen adult-vocal D are plotted in descending absolute magnitude.

Summary: - positive contribution sum = +3.746740; - negative contribution sum = -5.172237; - 15 positive / 13 negative dimensions; - cancellation ratio = 0.840173.

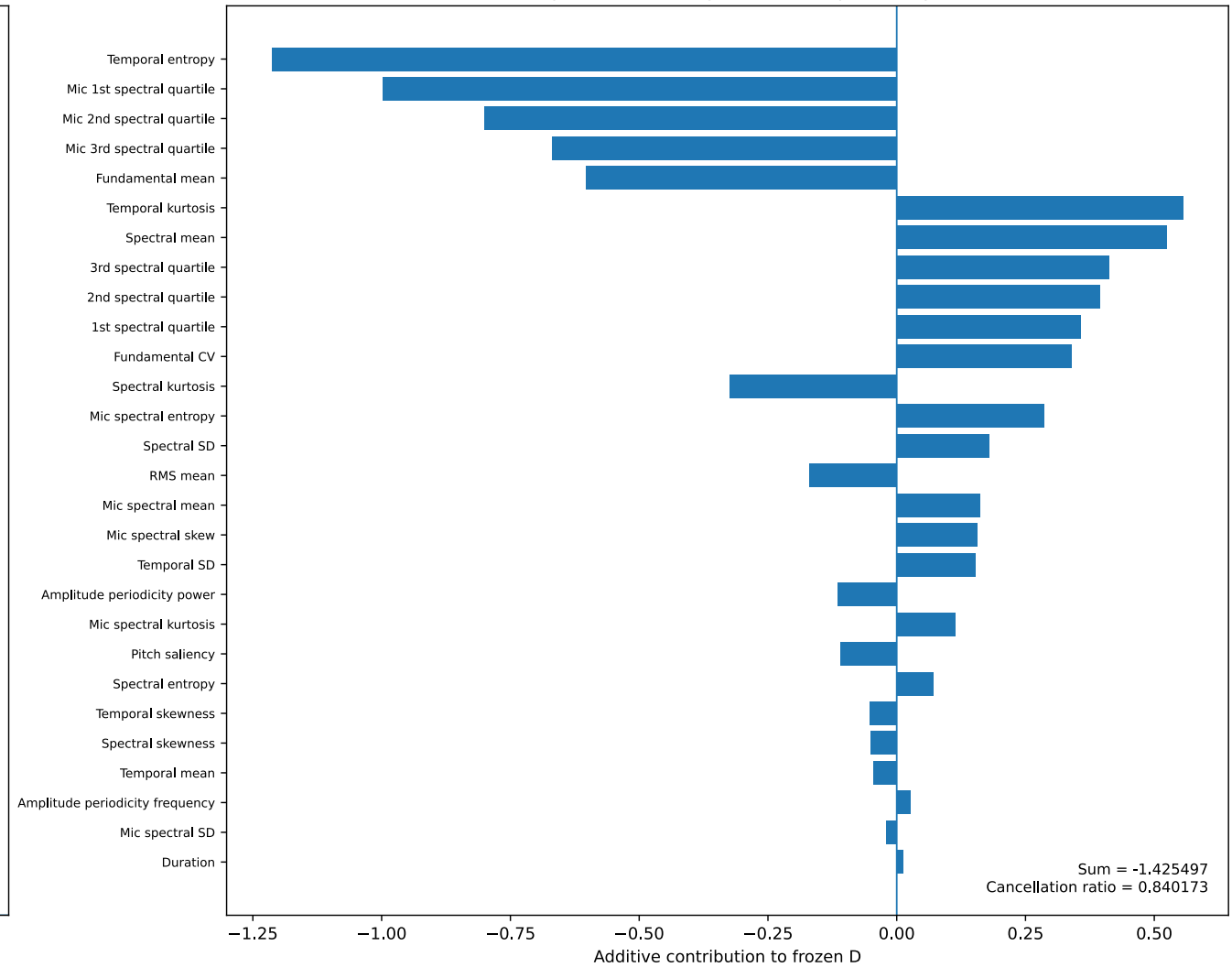
The contribution vector sums exactly to the frozen primary D=-1.425497. No feature-wise p-values or reduced-feature subset are used.

See Supplementary Material Figure S2.

A. Rachum enrichment
3/3 contributions positive; descriptive only



B. Elie auditory-feedback decomposition
15 positive / 13 negative; descriptive only



Supplementary Material Figure S2. Developmental descriptive decompositions.

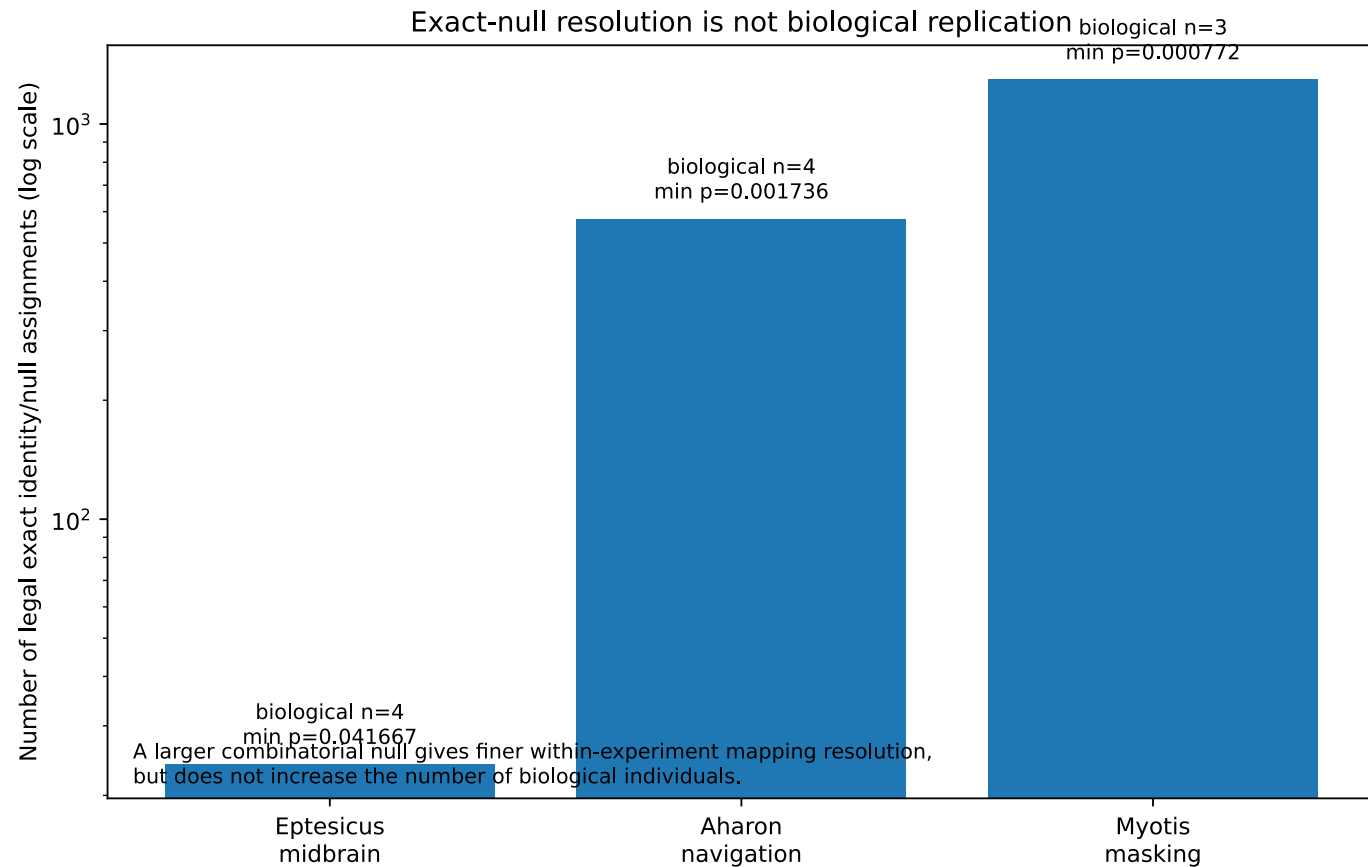
Supplementary Figure S3. Exact-null resolution and biological n

Exact identity/null spaces differ among the small-n controlled experiments:

- Eptesicus: 4 biological individuals, $4!=24$ identity mappings, minimum attainable exact one-sided $p=1/24=0.041667$;
- Aharon: 4 biological individuals across two independently permuted non-anchor conditions, $(4!)^2=576$ mappings, minimum $p=1/576=0.001736$;
- Myotis: 3 biological individuals under the frozen five-context architecture, 1,296 legal assignments, observed true mapping uniquely most extreme.

The size of the combinatorial null quantifies resolution of the **within-experiment identity-mapping test**. It does not increase the number of biological individuals and must not be interpreted as population-level replication.

See Supplementary Material Figure S3.



Supplementary Material Figure S3. Exact-null resolution and biological sample size.

Supplementary Methods S1. Evidence-tier rules

Define: - frozen primary; - predeclared secondary; - post-primary diagnostic; - descriptive/exploratory result; - structural STOP; - failed frozen gate.

State explicitly that a lower evidence tier cannot rescue a failed higher-tier gate.

Supplementary Methods S2. Cross-study synthesis boundary

State: - no pooled quantitative meta-analysis; - no shared effect-size scale across endpoints; - common inferential object for acute perturbation = same-individual correspondence after shared context effects are removed; - source discovery and cross-study synthesis were iterative; - source-level endpoints/nulls were frozen before outcome opening.

Supplementary publication-overlap disclosure

Several public tracking datasets used as a downstream wild-boundary analysis also appear in a separate vertical-individuality manuscript. The synthesis does not count those vertical-shape results as independent positive evidence. The older Tadarida-only Movement Ecology draft containing the same V1/V2 analysis now incorporated into the vertical-individuality manuscript has been retired from separate submission.